

DANIEL MIHOVCH

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Education

University of Delaware

Newark, DE

Bachelor of Science in Computer Science, conc. Systems & Networks

Aug 2023 – May 2027 (Expected)

Experience

University of Delaware: CIS - Summer Scholars

Newark, DE

Undergraduate Researcher

Jun 2026 – Aug 2026

- Profiled activation outliers at 73 sites across ViT-B/16 and measured accuracy impact via surgical zeroing ablation. Per-channel thresholds preserve +3.76 pp top-1 over global thresholds (95% CI significant)
- Implemented Pébay (2008) parallel higher-moments merge for exact streaming statistics (mean/variance/skewness/kurtosis) without materializing full activation tensors
- Traced per-channel variance patterns to their architectural origin: the interaction between the first MLP projection weights and the preceding LayerNorm scales (Pearson $r = 0.75\text{--}0.77$ in late encoder blocks), showing outliers are structurally encoded, not noise

Qnity Electronics

Wilmington, DE

Software Engineering & AI Enablement Intern

May 2026 – Aug 2026

- Created a Python data cleaning pipeline to reconcile company HR data with M365 Copilot usage metrics, resolving errors across both datasets to form one cohesive, truthful dataset covering 7,000+ employees and contractors with over 99.8% accuracy
- Developed multiple web-based dashboards for visualizing M365 Copilot usage and adoption metrics across the organization & drove enterprise-wide AI adoption by running demo and training sessions for hundreds of employees globally

FLIE Labs - Vertically Integrated Projects: Drone

Newark, DE

Software Engineer

Feb 2026 – Present

- Implementing autonomous swarming logic for DJI Matrice 100 drones utilizing ROS
- Developed a custom SITL simulator environment for DJI Matrice 100 swarms (Gazebo, ROS)
- Implementing an online, 3D first person interactive tour of the University of Delaware campus utilizing WebGPU and mesh models constructed from drone footage using 3D Gaussian Splatting

University of Delaware CIS

Newark, DE

Systems Programming Teaching Assistant

Aug 2024 – Present

- TA for CISC106 (Python for Engineers), CISC210 (Systems Programming) & CISC260 (Computer Architecture & ARMv8 Assembly)
- Heavily assisted in running the CISC210 lecture section, including delivering full lectures to the class
- Led weekly practicum sessions for 25+ students; held office hours and graded assignments
- Taught the C programming language, ARMv8 assembly, Bash, Unix systems, and Raspberry Pi hardware interaction

Four Youth Productions

Wilmington, DE

STEM Instructor & Studio Operations Assistant

Feb 2025 – Present

- Designed & led after-school STEM lessons (environmental science, physics, CS, chemistry) for 300+ underrepresented students across four Wilmington, DE public schools, grades Pre-K through 8th
- Assisted in managing community events to showcase student work

Projects

Piemulator - Raspberry Pi HAL | C, ASM, Bash, Raspberry Pi

Nov 2025 – Jan 2026

- Created a Hardware Abstraction Layer that serves as a drop-in replacement for the University of Delaware's *libsense* Raspberry Pi Sense Hat library
- Focused on portability and ease of use, currently only depends on libc and pthreads. Uses ANSI escape codes for terminal rendering
- Students can test and develop projects without always needing access to the physical Raspberry Pi, simply needing to link against my libpie instead of the provided libsense

DanOS - 16 bit Virtual Machine | C, Computer Architecture

Aug 2025 – Nov 2025

- Single tasking, 16 bit virtual machine written in C
- Implemented my own RISC-style ISA featuring a segmented memory model and multiple addressing modes

Technical Skills

Languages: C, C++, Rust, Bash, Python, Typescript

Frameworks/Libraries: CUDA, Raylib, SDL3, ncurses, ROS, WebGPU, React, Svelte